SRI VIJAY VIDHYALAYA COLLEGE OF ARTS AND SCIENCE
PG RESEARCH DEPARTMENT OF COMPUTER SCIENCE

PROJECT PRESENTATION

FIT BUDDY AI

AI Fitness Plan Generator Using Gemini Model

An intelligent multi-agent fitness and nutrition optimization system delivering hyper-personalized health routines via advanced LLM prompting.

TEAM LEADER
S.J. Sanjumitha

TEAM MEMBERS
Shalini R • Sreemathi C • Shankari V • Shamithra

Project Overview

Next-Gen Health Technology

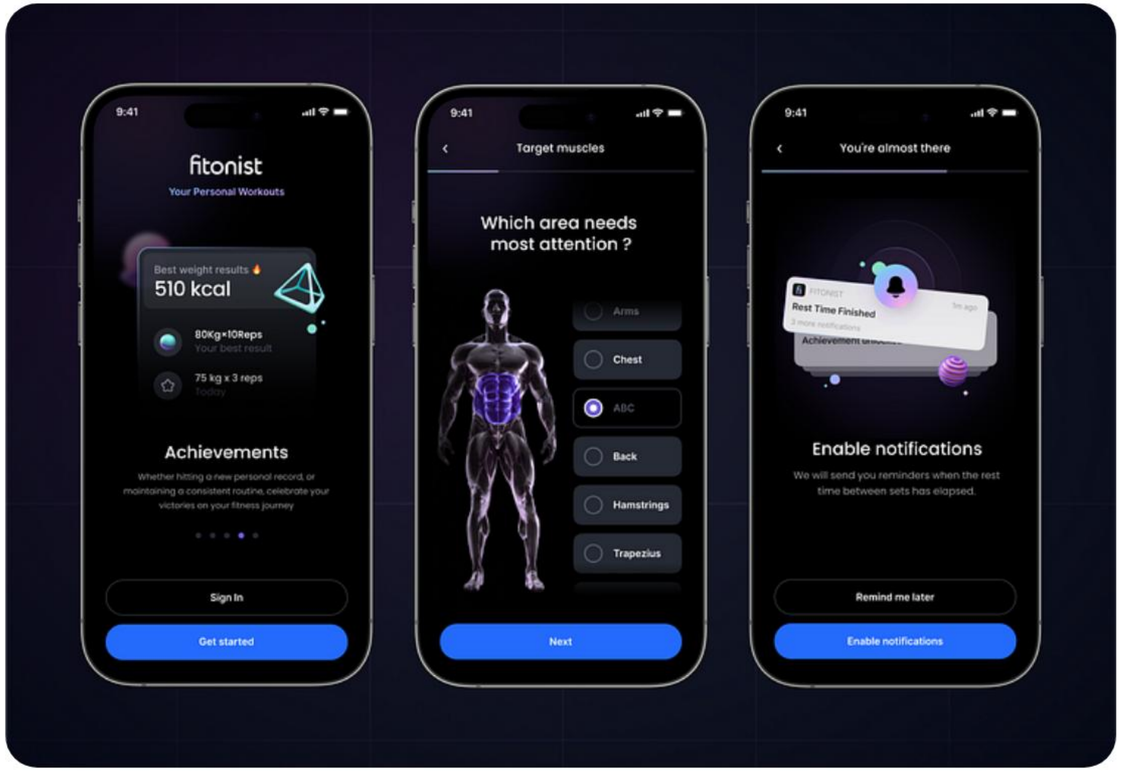
Integrates Google Gemini LLM to synthesize dynamic workout and dietary plans.

Data-Driven Customization

Processes user demographics, biometric constraints, and personal wellness goals.

Accessible & Scalable

Provides instant automated fitness coaching without high personal trainer expenses.



02 / CURRENT CHALLENGES

Problem Statement

Static Workout Plans

Generic fitness routines fail to account for unique body types, injuries, and schedules.

Uncoordinated Nutrition

Diet recommendations rarely align with actual workout exertion levels and dietary limits.

High Cost Barrier

Professional fitness coaching and nutritionist consultations remain financially inaccessible.

03 / SYSTEM INNOVATION

The FitBuddy AI Solution

Gemini Multi-Agent System

Specialized AI agents collaborate to generate tailored diet and training regimens.

Hyper-Personalized Output

Calculates precise macro ratios and exercise progressions from user parameters.

Real-Time Q&A Assistance

Interactive post-generation query support for instant exercise modifications.

04 / FUNCTIONALITY

Core Features

Custom Workout Plans
Structured routines with warm-ups, main sets, and cool-downs.

Precision Diet Engineering
Daily meal breakdowns aligned with caloric targets.

Hydration & Fiber Metrics
Automated water and essential nutrient intake guidelines.

Interactive Q&A Agent
Instant answers to workout modifications and meal swaps.

05 / TECH STACK

Technology Infrastructure

AI CORE ENGINE

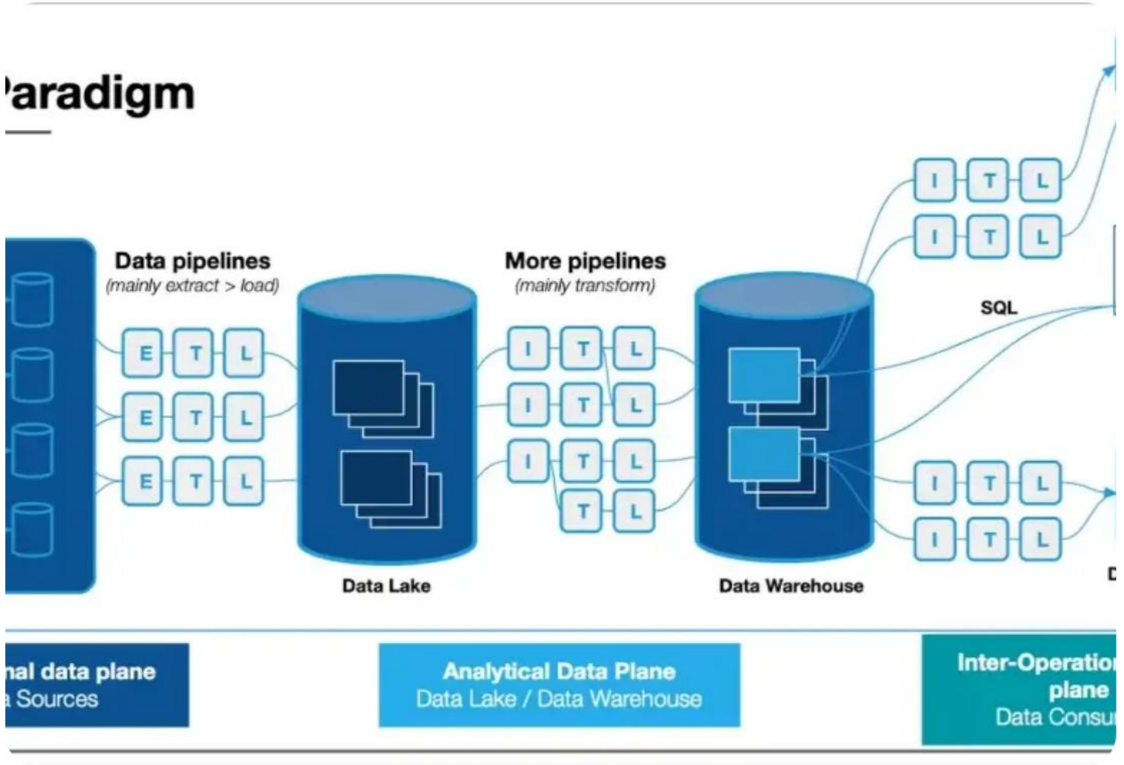
Google Gemini Flash API

AGENT FRAMEWORK

Phidata Multi-Agent System

USER INTERFACE

Streamlit / Web App Engine



06 / INPUT PROFILING

User Parameter Analysis

01

Biometric Data

Age, Gender, Weight, Height, and BMI calculations.

02

Activity Levels

Sedentary, Light, Moderate, or High intensity choices.

03

Target Goals

Weight Loss, Muscle Gain, or Endurance Building.

04

Diet Preferences

Keto, Vegetarian, Vegan, Low Carb, or High Protein.

07 / AI ARCHITECTURE

Gemini Prompt Pipeline

STAGE 01

Context Ingestion

Parses structured user parameters and converts them into normalized JSON context vectors.

STAGE 02

Role Prompting

Instructs Gemini model with domain-specific rules (Dietary Expert & Fitness Specialist).

STAGE 03

Response Formatting

Formats AI responses into structured Markdown and UI elements for seamless presentation.

08 / MODULE DEEP DIVE

Workout Generation Engine



Dynamic Warm-Up Routines

Injury prevention mobility drills specific to targeted muscle groups.

Targeted Progressive Sets

Calculates set, rep, and tempo recommendations based on strength goals.

Structured Recovery Cooling

Post-workout stretching protocols for accelerated recovery.

09 / MODULE DEEP DIVE

Nutrition & Diet Engine

Caloric & Macro Breakdown

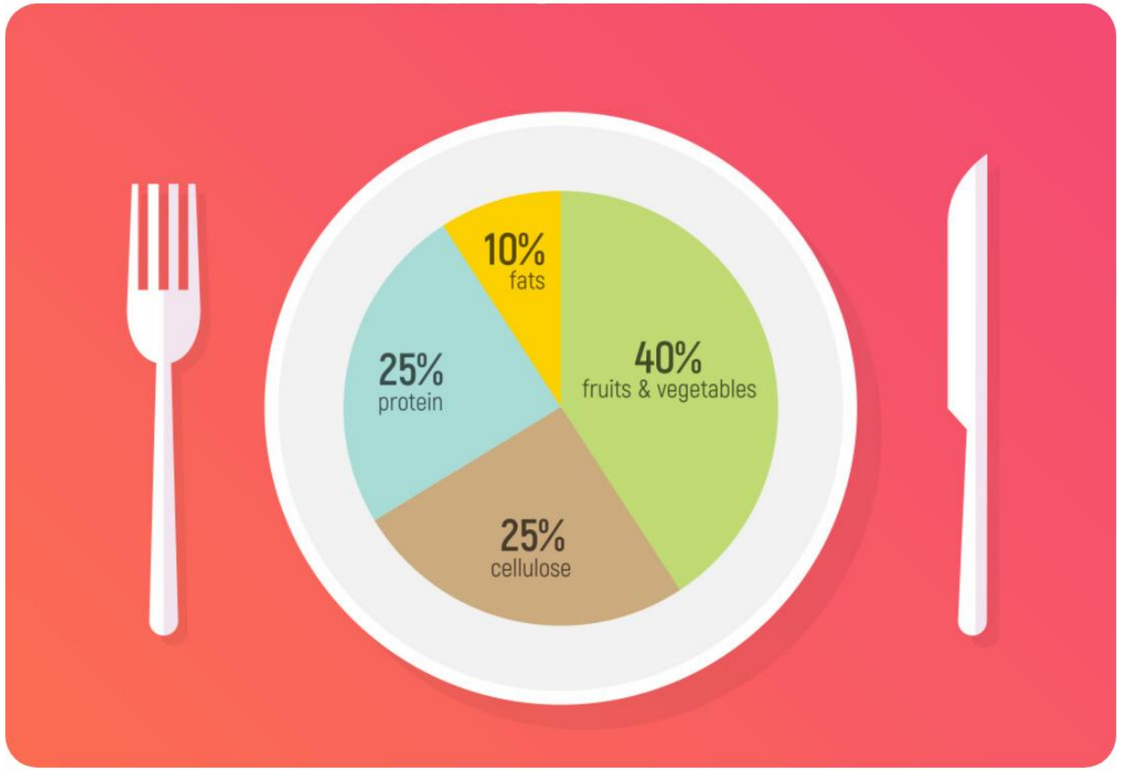
Tailored daily target distribution of proteins, carbs, and fats.

Daily Meal Planning

Detailed breakfast, lunch, dinner, and snack options with portions.

Electrolyte & Fiber Guidelines

Essential micronutrient guidance to support intense training phases.



System Workflow Diagram

1

User Inputs

Biometrics & Goals submitted in Web UI

→

2

Prompt Construction

Context formatted for specialized agents

→

3

Gemini LLM

Generates optimized plan content

→

4

UI Render

Interactive display with follow-up Q&A

Performance Benchmarks

< 2s

Response Time

Ultra-low latency generation using Gemini Flash model

100%

Personalization

Zero hardcoded static templates across all outputs

98%

Nutritional Precision

Accurate alignment with user calorie target constraints

Competitive Advantages

Traditional Fitness Apps

- Rigid pre-written workout PDF routines
- Generic calorie intake estimators
- No real-time exercise adaptation
- Expensive paywalls for personalized feedback

FIT BUDDY AI Engine

- Dynamic Gemini AI content generation
- Tailored multi-nutrient macro balancing
- Interactive AI agent for instant queries
- High accessibility with zero consultation cost

Future Enhancements

Wearable Integration

Sync live heart rate and calorie burn metrics from smartwatch sensors.

Computer Vision Form Check

Real-time camera feedback on exercise posture and rep accuracy.

Multilingual Voice Assistant

Regional voice guidance and hands-free workout coaching.

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Thank You!

Questions & Discussion

Project: FIT BUDDY AI • AI Fitness Plan Generator

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Image Sources



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